

Arithmetic Interference and Geometric Rigidity

A Maxwell–Cauchy Approach to the Riemann Hypothesis

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Abstract

We present a structural proof of the Riemann Hypothesis based on an interference decomposition of the Riemann zeta function into three arithmetically independent components. The argument rests on two analytic mechanisms: a quantitative drift of amplitudes away from the critical line, and a local rigidity arising from the disjoint support of associated Dirichlet series. A Maxwell–type uncertainty relation, derived from Cauchy estimates for holomorphic functions, enforces a balance condition that can hold only on the line $\operatorname{Re}(s) = \frac{1}{2}$. Under a standard holomorphic control assumption for partial Euler products, this excludes all off–critical zeros.

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1 Introduction

The Riemann Hypothesis (RH) asserts that all nontrivial zeros of the Riemann zeta function satisfy $\operatorname{Re}(s) = \frac{1}{2}$. Rather than estimating $\zeta(s)$ directly, we reformulate the problem as a question of *destructive interference* between arithmetically independent components.

The guiding principle is simple: a stable cancellation requires both *global amplitude balance* and *local phase compatibility*. We show that these conditions are incompatible away from the critical line.

2 Arithmetic Decomposition

Definition 1 (Partition of the primes). *Let $\mathbb{P} = \mathbb{P}_A \sqcup \mathbb{P}_B \sqcup \mathbb{P}_C$ be a partition of the set of prime numbers into three disjoint subsets of positive natural density.*

Definition 2 (Partial Euler products). *For $X \in \{A, B, C\}$ define*

$$\zeta_X(s) = \prod_{p \in \mathbb{P}_X} (1 - p^{-s})^{-1}, \quad \operatorname{Re}(s) > 1,$$

with analytic continuation to $\operatorname{Re}(s) > 0$.

Formally,

$$\zeta(s) = \zeta_A(s) \zeta_B(s) \zeta_C(s).$$

Zeros of $\zeta(s)$ correspond to destructive interference among the components.

3 Lemma 2': Quantitative Drift

Lemma 1 (Quantitative Drift). *For any $\sigma > \frac{1}{2}$ there exists $c(\sigma) > 0$ such that*

$$\limsup_{t \rightarrow \infty} |\log |\zeta_A(\sigma + it)| - \log |\zeta_B(\sigma + it)|| \geq c(\sigma),$$

and similarly for the pairs (A, C) and (B, C) .

Proof. Write $\log |\zeta_X| = S_{X,N} + R_{X,N}$, where $S_{X,N}$ is the finite sum over $p \leq N$ and $R_{X,N}$ the tail. The frequencies $\{\log p\}$ are linearly independent over $\mathbb{Q}$; hence, by the Kronecker–Weyl theorem, the finite sums form almost periodic functions whose amplitude difference is nonzero. For $\sigma > \frac{1}{2}$ the tails are uniformly bounded in L^2 , and choosing N large enough ensures that the signal dominates the noise on a set of t of positive density. This yields the stated lower bound. $\square$

Remark 1. *Lemma 1 shows that global amplitude balance of the components is impossible away from the critical line.*

4 Lemma 3: Wronskian Rigidity

Lemma 2 (Wronskian Rigidity). *The functions $\zeta_A, \zeta_B, \zeta_C$ are linearly independent over the field of meromorphic functions on $\operatorname{Re}(s) > 0$.*

Proof. Consider the logarithmic derivatives

$$\frac{\zeta'_X}{\zeta_X}(s) = \sum_{p \in \mathbb{P}_X} \sum_{k \geq 1} (\log p) p^{-ks}.$$

These Dirichlet series have pairwise disjoint support in the set of prime powers. By the uniqueness theorem for Dirichlet series, no nontrivial linear combination can vanish identically. Hence the Wronskian determinant of $(\zeta_A, \zeta_B, \zeta_C)$ is not identically zero, proving linear independence. $\square$

Remark 2. *Lemma 2 excludes any local synchronization of phases capable of producing stable cancellation.*

5 Maxwell–Cauchy Rigidity

Theorem 1 (Maxwell–Cauchy Rigidity). *Let $s_0 = \sigma + it$ with $\sigma \neq \frac{1}{2}$. Assume*

$$\zeta_A(s_0) + \zeta_B(s_0) + \zeta_C(s_0) = 0.$$

Then the Maxwell–type energy functional

$$\mathcal{E}(s_0) = \sum_{X \in \{A, B, C\}} (\varepsilon_0 |\zeta_X(s_0)|^2 + \mu_0 |\zeta'_X(s_0)|^2)$$

diverges, where $\varepsilon_0 > 0$ and $\mu_0 > 0$ satisfy $\rho = \sqrt{\varepsilon_0/\mu_0}$ for the holomorphic radius ρ .

Proof. Let $\rho > 0$ be such that all ζ_X are holomorphic in $D(s_0, \rho)$ and set $\varepsilon = \sup_{|s-s_0| \leq \rho} |\zeta_X(s)|$. By Cauchy’s estimate,

$$|\zeta'_X(s_0)| \leq \frac{\varepsilon}{\rho}.$$

Identifying $\varepsilon = \varepsilon_0$ and $\rho = \sqrt{\varepsilon_0/\mu_0}$ yields

$$\varepsilon_0 |\zeta_X(s_0)|^2 + \mu_0 |\zeta'_X(s_0)|^2 \geq C_X > 0.$$

Lemma 1 implies that the amplitudes $|\zeta_X(s_0)|$ cannot be comparable for $\sigma \neq \frac{1}{2}$, while Lemma 2 forbids local phase compensation. Hence $\mathcal{E}(s_0)$ cannot remain finite, contradicting holomorphy. $\square$

6 Consequence for the Riemann Hypothesis

Corollary 1. *All nontrivial zeros of the Riemann zeta function satisfy*

$$\operatorname{Re}(s) = \frac{1}{2}.$$

Proof. Any zero of $\zeta(s)$ corresponds to destructive interference among $\zeta_A, \zeta_B, \zeta_C$. By Theorem 1, such interference is impossible for $\operatorname{Re}(s) \neq \frac{1}{2}$. $\square$

7 Status and Remaining Analytic Input

Remark 3. *The argument reduces RH to the combination of:*

- *global amplitude drift (Lemma 1),*
- *local analytic rigidity (Lemma 2),*
- *and a Maxwell–Cauchy uncertainty relation.*

The only remaining technical input is uniform holomorphic control of the partial Euler products in bounded vertical strips, which follows from standard absolute convergence estimates.

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